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Advances in chloride additives for high-efficiency perovskite solar cells: multiple points of view

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Chloride (Cl) additives are rather effective in improving the performance of perovskite solar cells (PSCs) through the modulation of crystallization process and surface morphology. After incorporating Cl-containing additives, the optoelectrical properties of perovskite films, such as the electron/hole diffusion length and carrier lifetime, are greatly enhanced. However, only a trace amount of Cl has been identified in the resultant perovskite film, and the mechanism of efficiency improvement induced by Cl remains unclear. In this review, we discuss organic and inorganic Cl additives systematically from the perspective of their solubility, volatility, cation size and chemical groups. In addition, the roles of residual Cl anions and cations are analyzed in detail. Finally, some valuable future perspectives of Cl additives are proposed.

1. Introduction

In the year 2023, the power conversion efficiency (PCE) of organic–inorganic metal halide perovskite solar cells (PSCs) hit a milestone of 26.1%,¹ which is comparable to that of the record of their silicon competitors (26.81%, HIT, LONGI, 2022–11). Along with their low cost and solution processing capability,^{2,3} PSCs have shown a rather bright industrial

commercialization possibility in the last 2–3 years.⁴ At least more than ten companies, for example the Oxford PV (UK), GCL (China), Microquanta Semiconductor (China), Saule Technologies (Poland) and UtmoLight (China), have initiated the fabrication of square meter sized PSC panels. It is well known that the optoelectronic and morphological properties of the perovskite light absorbing layer are key factors that determine the performance of large-area solar panels.^{5,6} However, controlling the nucleation and crystal growth over a large area is one of the biggest challenges during the fabrication process.⁷ Chloride (Cl) additives, known as “additive removal” agents,⁸ have been proved to be rather effective in crystal growth modulation.^{9–11} It was observed that the perovskite layer with a volatile Cl additive, for instance methylammonium chloride (MACl), had a ten times longer electron diffusion length,¹² ~1 μ s carrier

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lifetime,¹³ and μm scale grain size.¹⁴ Furthermore, less current-voltage hysteresis was also confirmed in PSCs with Cl additives.^{15,16} Surprisingly, only a trace amount of Cl could be identified in the final perovskite film after the annealing process. Therefore, it is widely accepted that the Cl additives function as intermediate phase modulation agents and get released. However, exact mechanism of Cl additives in crystallization modulation is still under debate because of the complex chemical reactions occurred in the solution.^{17–20}

Our review begins with the analysis of surface morphology and optoelectronic improvement induced by Cl additives, and then the mechanism of crystal phase modulation is discussed in the second section. In the third section, we summarized the methods that are frequently used in additive engineering. The organic and inorganic Cl additives were discussed extensively from the viewpoint of their solubility, volatility, cation size and chemical groups. To help the readers to understand the roles of various Cl additives, we provided a general framework as shown in Fig. 1. The solubility of Cl additives in typical PSC solution, such as dimethylsulfoxide (DMSO) and dimethylformamide (DMF), is the first and most important factor to begin with. A low solubility usually leads to phase segregation. The boiling point of the organic Cl additives is the second factor to be considered. Note that the boiling point in our discussion is related to the cation part for the organic Cl additives. For instance, the cation part of MACl is methylamine (MA^0), which has a boiling point of $-6.3\text{ }^\circ\text{C}$. While the boiling point of inorganic Cl additives is the traditional one that readers are familiar with. In general, the volatile low boiling point additives will get released after the annealing process, while the cations of nonvolatile additives will be left in the perovskite film. The size and chemical groups of the residual cations then come into play in the modulation of perovskite crystal structures.⁸ They can be incorporated into the crystal lattice, occupy the formamidinium (FA) or methylammonium (MA) sites, accumulate at the grain boundary or assemble at the layer interface. In general, the cations left behind could immobilize the excess iodine (I^-) anion, change the crystal tolerance factor (t), suppress the ion migration, passivate defects, release strain and facilitate two-dimensional (2D) phase formation.²¹ Table 1

summarises the improvement of the solar cell parameters after implementation of Cl additives. In most cases, the Cl additives improve the short-circuit current density (J_{sc}), open-circuit voltage (V_{oc}) and fill factor (FF) simultaneously. Only in limited cases, such as the ammonium chloride (NH_4Cl) additive with poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) as a hole transportation layer, the V_{oc} gets deteriorated, but the overall PCE is increased due to increased J_{sc} and FF.²²

2. Morphology, optoelectronic properties and mechanism

The efficiency of PSCs is highly dependent on the quality of the perovskite film, and more specifically, the surface morphology and optoelectronic properties. A compact surface and large crystal grain size are the prerequisites to obtain high performance. Tiny pin holes are not allowed because the electron transport layer (ETL) will contact with the hole transport layer (HTL) directly. Although a small grain size does not deteriorate the V_{oc} seriously,²³ the J_{sc} is greatly decreased due to the defects located at the grain boundaries (GBs). Furthermore, the J - V hysteresis and long-term stability issues arise because of the GBs, which then decrease the FF and leads to low efficiency.

2.1. Surface morphology

Almost all the Cl additives have the same effect on the surface morphology, that is, the crystal grain size is increased. It is observed that the organic Cl additives are more effective than that of the inorganic alternatives based on the crystal grain size, which is compared in Fig. 2b (MACl as additive) and Fig. 2f (CsCl as additive). However, the crystal grain size is smaller when Cl additives are absent regardless of the organic (Fig. 2a) and inorganic type (Fig. 2e). The possible reason behind the different effects of organic and inorganic Cl additives is their solubility in DMF/DMSO solution. For example, CsCl has a solubility of $<2\text{ mg mL}^{-1}$ in neat DMF,²⁷ while the MACl is highly soluble in typical DMF/DMSO solution. The release rate of the Cl additives is important for the crystallization modulation. For example, Seok *et al.* added the higher boiling point propylammonium chloride (PA^0 , $47.8\text{ }^\circ\text{C}$) into the 35 mol% MACl base (MA^0 , $-6.3\text{ }^\circ\text{C}$).²⁵ The distinct GBs were observed in the MACl perovskite film (Fig. 2c, top panel); however, when 15 mol% polyaluminum chloride (PACl) was added, the GBs disappeared (Fig. 2d, top panel). These results indicate that the intermediate phase induced by organic Cl additives can be finely tuned through their boiling point, in such a way the nucleation rate can be greatly suppressed. The *in situ* grazing-incidence wide-angle X-ray diffractions (GIWAXD) were collected for the perovskite films during heating from room temperature (RT) to $120\text{ }^\circ\text{C}$. The initial $\delta\text{-FAPbI}_3$ phase was converted into an $\alpha\text{-FAPbI}_3$ phase at the temperature of $60\text{--}80\text{ }^\circ\text{C}$ (Fig. 2c, middle panel), which is due to the partial exposure to moisture and has been observed in other studies.^{28,29} For the 15 mol% PACl added sample, there was a coexistence of $\alpha\text{-FAPbI}_3$ and $\delta\text{-FAPbI}_3$ even at RT (Fig. 2d,



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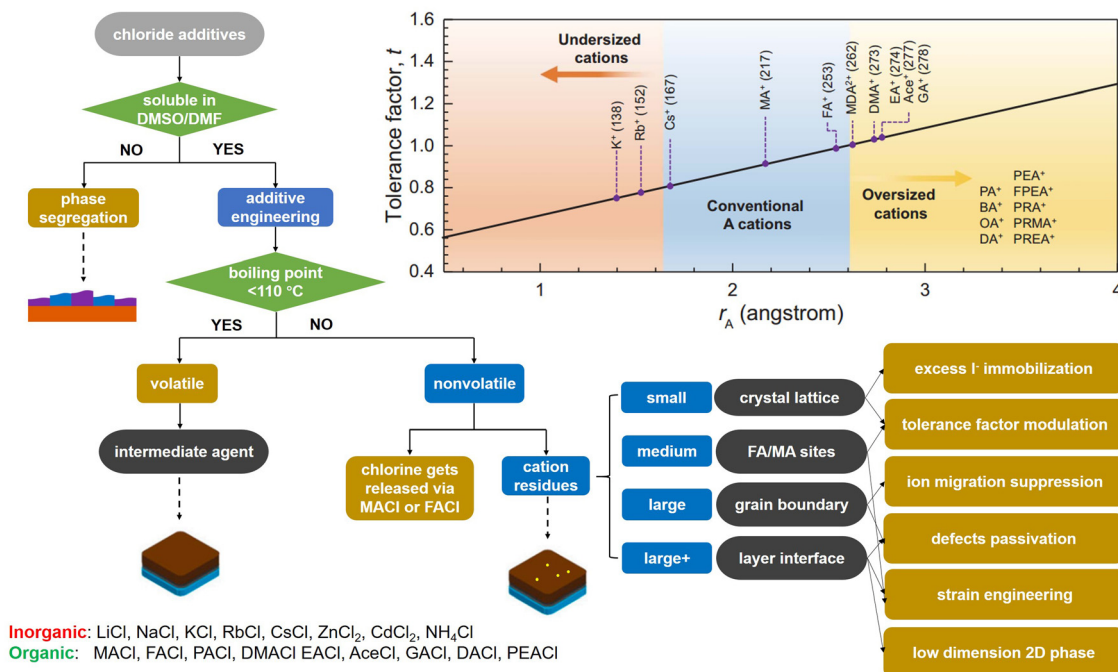


Fig. 1 The general framework of organic and inorganic Cl additives for FA/MA based PSCs from the viewpoint of their solubility, boiling point, volatility, cation size and chemical groups. Inset, the size of typical cation and the tolerance factor. Reproduced with permission.⁸ Copyright, 2022 AAAS.

middle panel). 2D GIWAXD of the perovskite thin films after annealing at 120 °C for 30 min showed that the crystal grain had a random orientation in the MACl perovskite film (Fig. 2c, bottom panel), while the crystal grain possessed a preferred orientation along the (100)_x direction for the 15 mol% PACl added perovskite film (Fig. 2d, bottom panel). Extremely flat perovskite film with PACl additive leads to a PCE of 26.08% (Fig. 2g), while the performance of cells with CsCl additive has a moderate PCE of 20.13% (Fig. 2h). Controlling the release rate of organic Cl additives thus can lower the crystal growth rate, which subsequently facilitate the preferred growth of grain in a specific direction.

Solubility and boiling point of organic Cl additives are two critical factors that determine the δ -phase to α -phase transition. First, a high Cl⁻ anion content is required to effectively control the nucleation process. For example, 100 mol% MACl was employed to form α phase perovskite at RT.³⁰ Second, the release rate of the Cl⁻ anion controls the intermediate phase by way of boiling point, and the underlying mechanism will be discussed later.

2.2. Optoelectrical properties

Beyond the surface morphology improvement, another key advantage of the perovskite films with Cl additives is the long electron/hole diffusion length and carrier lifetime. Snaith *et al.* found that the charge carrier diffusion length of MAPbI_{3-x}Cl_x is ten times longer than that of their MAPbI₃ counterpart, reaching 1 μ m.¹³ Ultraviolet visible (UV-Vis) absorption spectra is frequently used to measure the band gap of perovskite films. As shown in Fig. 3a, there was no discernible difference between reference perovskite and the one with the FACl additive, which

means that the Cl did not change the band gap of perovskite. This is reasonable because they get released in the end. In general, the volatile low boiling point Cl additives, such as the MACl, FACl and PACl, are widely believed to play a role as intermediate phase modulation agents and get released after the annealing process. Photoluminescence (PL) and time-resolved PL are other commonly used methods to study the lifetime and dynamic behaviors of charge carriers. The perovskite film with FACl had a higher PL intensity under identical excitation conditions, and the time-resolved PL (TRPL) showed a longer decay time (Fig. 3b). David *et al.* studied the impact of microstructure on the local carrier lifetime in PSCs. Fig. 3c shows the overlaid scanning electron microscopy (SEM) micrograph and the fluorescence image. The PL intensity varied widely even between grains, even though no discernible difference could be observed for the grains. However, there is a strong correlation between the PL intensity and the content of Cl (Fig. 3d). One hypothesis is that during the release of Cl additives, the Cl anion migrates to the perovskite surface, which then passivates the defect trap state from the trace amount of the Cl residue.¹¹ Further studies in the future are required to understand the dynamic release process of Cl additives. The molecular dynamic simulation might be an efficient way to elucidate the role of Cl in perovskite films for electronic property enhancement.

2.3. Mechanism

The exact mechanism of Cl additives for PSC performance improvement is still under extensive debate. According to the volatile capability of the cations, the Cl additives can be grouped into two categories. One is the volatile cations, such

Table 1 Summary of PSCs with typical Cl additives

Chloride additives	Boiling point and size of cation	Device structure	J_{sc} (mA cm ⁻²)	V_{oc} (V)	FF	PCE (%)	Ref.
MA-Cl	217 pm −6.3 °C	FTO/SnO ₂ /MAPbI ₃ /spiro/Au	19.99 21.87	1.01 1.06	0.70 0.75	13.97 17.22	24
MA-Cl	217 pm −6.3 °C	FTO/c-TiO ₂ /m-TiO ₂ /FAPbI ₃ /spiro/Au	24.84 25.92	1.027 1.130	0.77 0.82	19.66 24.02	34
FA-Cl	253 pm 46.3 °C	FTO/c-TiO ₂ /CsFAMAPbI _{3-x} Br _x /spiro/Au	22.53 23.63	1.152 1.183	71.79 75.21	18.60 21.02	62
FA-Cl	253 pm 46.3 °C	FTO/SnO ₂ /FAMAPbI _{3-x} Br _x /spiro/Au	22.48 23.01	1.110 1.134	0.778 0.785	21.2 22.12	31
MDA-Cl ₂	262 pm 82.6 °C	FTO/c-TiO ₂ /m-TiO ₂ /FAPbI ₃ /spiro/Au	25.14 26.50	1.14 1.14	0.8055 0.8177	23.05 24.66	63
DMA-Cl	273 pm 7 °C	FTO/SnO ₂ /perovskite/spiro/Au (n-i-p) FTO/PTAA/m-Al ₂ O ₃ /perovskite/C ₇₀ /Au (p-i-n)	21.3 21.0 19.9 21.0	1.12 1.19 1.06 1.06	0.79 0.80 0.74 0.74	18.8 20.2 16.2 16.7	42
EA-Cl	274 pm 240 °C	FTO/c-TiO ₂ /m-TiO ₂ /MAPbI ₃ /spiro/Au	22.35 23.51	1.03 1.08	0.7301 0.7699	17.35 20.03	64
Ace-Cl	278 pm 62.8 °C	ITO/SnO ₂ /MAPbI ₃ /PCBM/Ag	20.67 24.46	0.914 1.013	0.64 0.73	12.10 18.12	65
GA-Cl	278 pm 132.9 °C	ITO/PTAA/MAPbI ₃ /C ₆₀ /Al	19.62 21.32	1.11 1.13	0.7191 0.7905	15.66 18.65	66
PA-Cl	380 pm 47.8 °C	FTO/SnO ₂ /FAPbI ₃ /spiro/Au	25.6 25.69	1.175 1.178	0.83 0.8615	24.96 26.07	25
DA-Cl	1516 pm 265 °C	ITO/SnO ₂ /FAPbI ₃ /spiro/Au	24.69 24.80	1.132 1.174	0.8101 0.8211	22.66 23.91	67
PEA-Cl	647 pm 196.5 °C	FTO/c-TiO ₂ /m-TiO ₂ /FAPbI ₃ /spiro/Ag	22.2 22.7	1.05 1.14	0.647 0.70	15.0 18.1	68
AI-Cl	742 pm 229 °C	ITO/NiO _x /PTAA/Al ₂ O ₃ /FAPbI ₃ /C ₆₀ /Ag	25.2 25.67	1.094 1.128	0.732 0.786	20.18 22.78	36
NH ₄ Cl	143 pm 520 °C	ITO/PEDOT:PSS/MAPbI ₃ /PC ₆₁ BM/Al	12.78 14.08	0.92 0.88	0.694 0.801	8.16 9.93	22
LiCl	600 pm 1382 °C	FTO/c-TiO ₂ /m-TiO ₂ /ZrO ₂ /(Li)MAPbI ₃ /C	17.55 20.27	0.878 0.927	0.64 0.77	9.9 14.4	43
KCl	138 pm 1420 °C	ITO/SnO ₂ /(K)FAPbI ₃ /spiro/Au	21.59 21.82	1.108 1.124	0.759 0.793	18.12 19.44	46
RbCl	146 pm 1390 °C	FTO/SnO ₂ /(Rb)FAPbI ₃ /spiro/Au	25.1 25.8	1.17 1.19	0.8376 0.8338	24.6 25.6	41
CsCl	167 pm 1297 °C	ITO/c-NiO/(Cs)FAPbI ₃ /C ₆₀ /Cu	22.71 23.54	1.027 1.064	0.78 0.80	18.06 20.13	26
ZnCl ₂	74 pm 732 °C	FTO/c-TiO ₂ /MAI(PbI ₂) _{1-x} (ZnCl ₂) _x /spiro/Au	21.09 22.04	1.05 1.09	0.7405 0.7576	16.4 18.2	40
CdCl ₂	96 pm 960 °C	FTO/c-TiO ₂ /m-TiO ₂ /MAPbI ₃ /spiro/Au	19.78 20.87	0.962 1.000	0.657 0.732	12.17 15.28	39

as the MA-Cl and FA-Cl. The main role of this class of Cl additives is intermediate phase modulation agent. In contrast, another class is the nonvolatile cation, such as cadmium chloride (CdCl₂) and phenethylammonium chloride (PEA-Cl). The role of these Cl additives becomes rather complex because of the size and chemical groups of the cation residues.

In situ PL measurements have been performed by Alex *et al.* to uncover the role of volatile Cl additives in the crystallization process of FAPbI₃.³³ As shown in Fig. 4a, pure FAPbI₃ requires a high annealing temperature of 150 °C and long time period of 1200 s to form the α -phase, and a PL intensity of 3×10^2 was obtained after the annealing process. However, adding NH₄Cl into the precursor solution could lower the temperature down to 130 °C, and a PL intensity of 1.3×10^4 was achieved after the annealing process, which indicates a higher crystallinity. The authors found that the volatile MA-Cl exhibited a similar effect. Furthermore, MA-Cl had an additional crystallization route, which is shown in the PL mapping. The maximum PL peak position of the MA-Cl-based sample gradually shifted from 700

to 780 nm during the first 100 s of annealing, suggesting an MA-assisted nucleation process. The temperature was further lowered down to 110 °C and a much higher PL intensity of 1×10^5 was obtained. Density functional theory (DFT) calculations have provided insightful clues for the intermediate phase from the viewpoint of formation energy (Fig. 4b). It was found that upon the incorporation of Cl into the crystal structure of FAPbI₃, the formation energy is lowered, and simultaneous incorporation of Cl and MA can further lower the formation energy. During the annealing process, I⁻ exchanges with the Cl⁻ and the MA is partially left behind in the crystal lattice.³⁴ It is calculated that when MA is left in the crystal lattice, the formation energy is further lowered, which means that the MA_xFA_{1-x}PbI₃ is more energetically stable. The calculations are consistent with the experimental findings in Fig. 4a, where the perovskite with MA-Cl additive had the lowest annealing temperature. Apart from the low bandgap perovskite, MA-Cl has been found to be effective in crystal growth modulation for wide bandgap perovskite, for instance the FA_{0.83}Cs_{0.17}Pb(I_{0.6}Br_{0.4})₃.

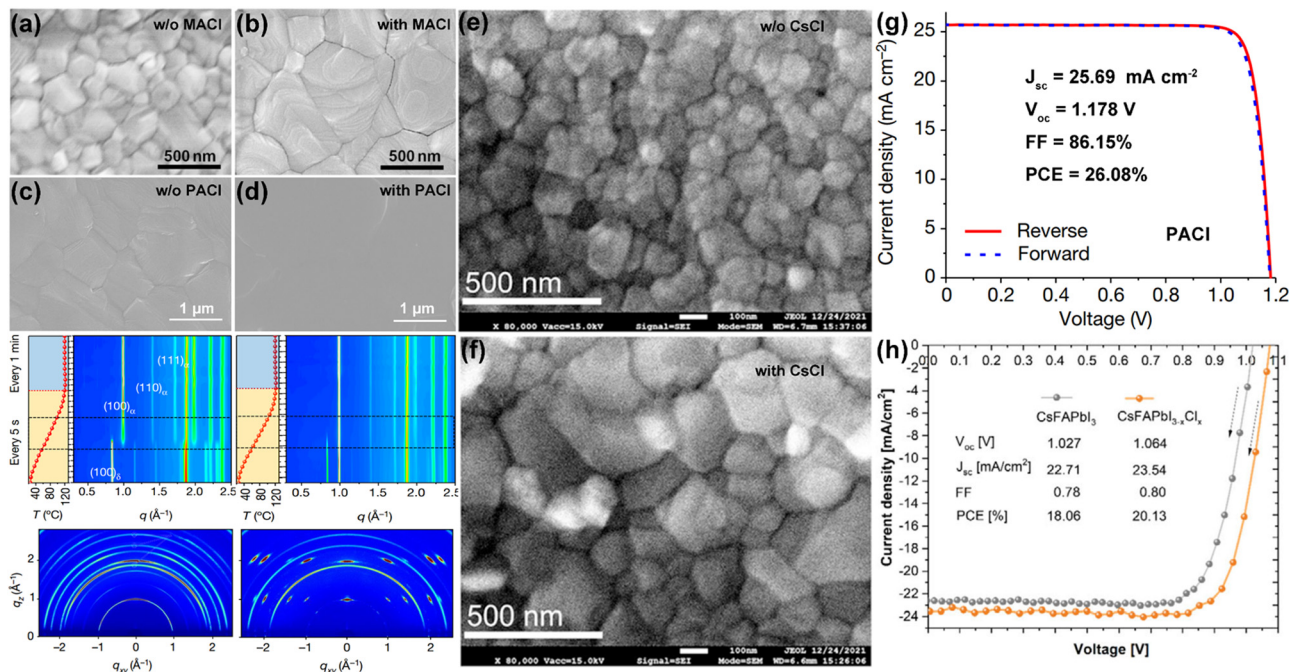


Fig. 2 Typical SEM images of the perovskite films with organic and inorganic Cl additives. (a) and (b) MACl additive with DMF as a solvent and ethyl acetate (EA) as an antisolvent. Reproduced with permission.²⁴ Copyright, 2016 Elsevier Ltd. Surface morphology, *in situ* GI-WAXD during heating from RT to 120 °C, and 2D GI-WAXD of FAPbI₃ without PACl (c) and with PACl (d) as a precursor additive. Reproduced with permission.²⁵ Copyright, 2023 Nature. SEM images of the perovskite films with (e) and without (f) CsCl additive. Reproduced with permission.²⁶ Copyright, 2023 Wiley-VCH. Performance of the PSCs with chloride additive PACl (g) and CsCl (h). Reproduced with permission.^{24,25} Copyright, 2023 Nature. Copyright, 2023 Wiley-VCH.

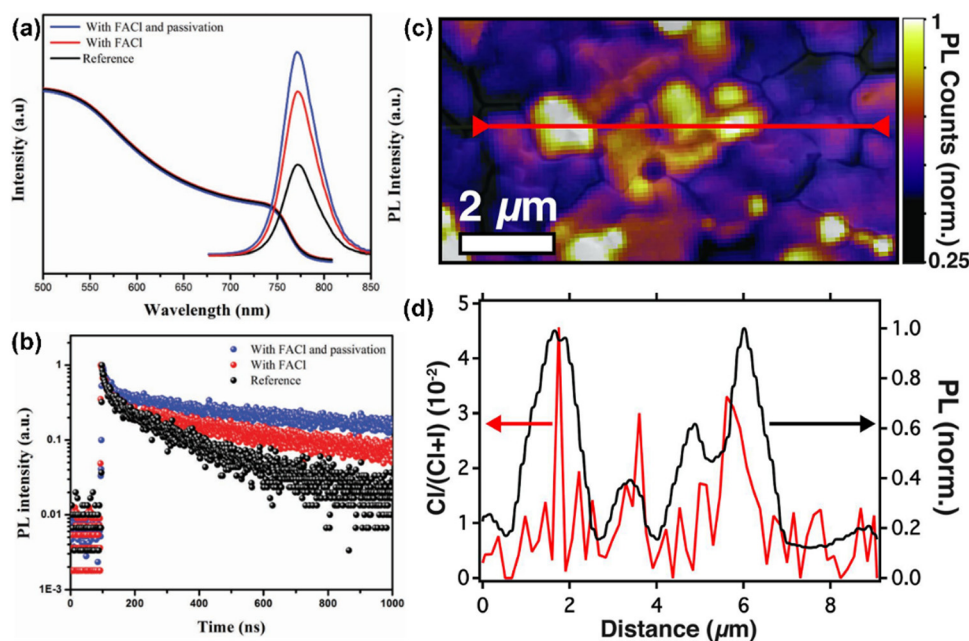


Fig. 3 UV-vis and PL spectra (a) and TRPL (b) curves of perovskite films without and with FACL additive. Reproduced with permission.³¹ Copyright, 2019 Wiley-VCH. SEM micrograph overlaid on the fluorescence image (c) and (d) EDS line scan showing the local elemental weight ratio of Cl/(Cl + I) tracks areas of higher integrated PL intensity, indicating that Cl is associated with better performing grains. Reproduced with permission.³² Copyright, 2015 AAAS.

Shen *et al.* found that addition of MACl results in Cl-rich phases initially, which is reflected in crystal lattice reduction that is

observed in XRD patterns (Fig. 4c);³⁵ complex intermediate poly-type phase mixture is formed shortly afterward, and finally evolves

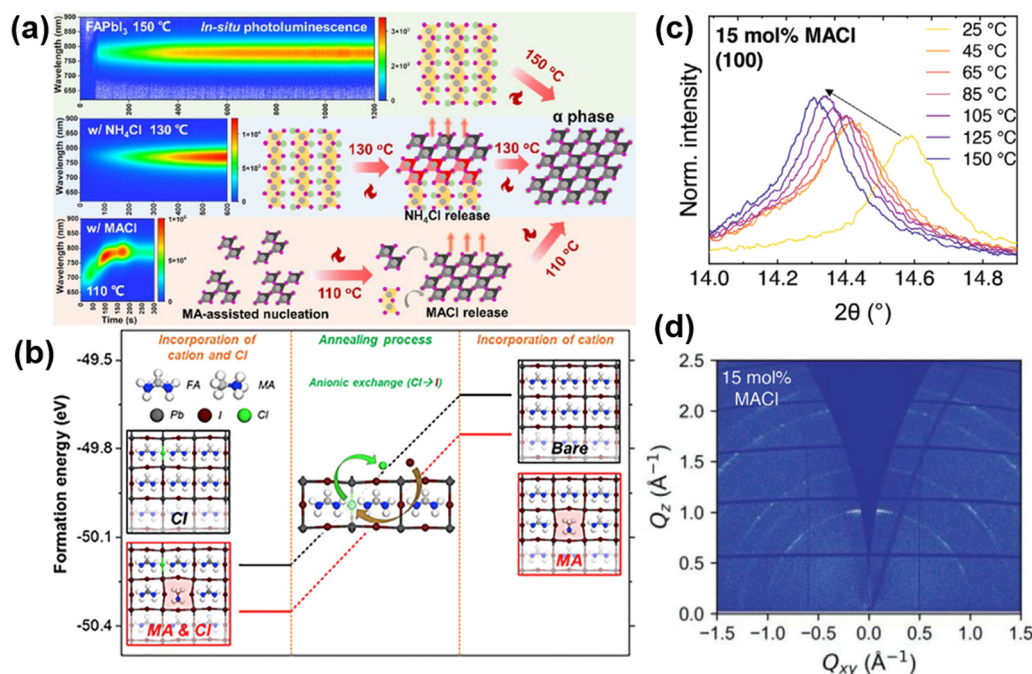


Fig. 4 (a) Crystallization mediated by typical volatile chloride additives NH₄Cl and MACl. Reproduced with permission.³³ Copyright, 2023 American Chemical Society. (b) Formation energy of α -FAPbI₃ with and without MACl. Reproduced with permission.³⁴ Copyright, 2019 Elsevier. (c) Normalized XRD intensity as a function of anneal temperature of wide bandgap perovskite with the 15 mol% MACl additive. (d) 2D GIWAXS scattering pattern of wide bandgap perovskite that is thermal treated after 250 s. Reproduced with permission.³⁵ Copyright, 2023 Wiley-VCH.

into the crystallization regime where the Cl is replaced gradually by the I and Br. Fig. 4d shows the GIWAXS scattering pattern, it shows that the perovskite (100) Debye-Scherrer ring is more intense and concentrated along the Q_z direction, this means the (100) planes of perovskites are oriented primarily out of plane, which indicates that initial perovskite nuclei with 15 mol% MACl are highly oriented corner-sharing crystallites.

Roles of the nonvolatile Cl additives are much more complex than that of their volatile counterparts. The Cl anion gets released during the annealing process in the form of FAPbCl or MACl. However, the cations are left behind as residues, which can dope the crystal lattice (potassium, K⁺ or rubidium, Rb⁺), accumulate at the GBs (PEA⁺), or form assembled contact layer on the surface (dodecylammonium, DA⁺). Roles of the cations are rather complex, such as immobilize the excess halide, tune the structure tolerance factor, interact with charge carriers, impede ion migration at the grain boundary, release crystal strain, passivate the defects, form hydrogen bonds, and align the energy levels. Fig. 5 illustrates the size and possible locations of cations, as well as the rich role of different cations. The main drawback of the nonvolatile Cl additives is that the added amount is generally relatively low, which means that the Cl anion content is low, and therefore the modulation of nucleation and crystal growth becomes weak. For example, in our recent study, it was found that the incorporation of 2-aminoindane hydrochloride (AICl) into the FAPbI₃ precursor solution only increased the crystal grain size slightly, but a powerful 2D/3D phase was formed at the buried interface due to the large size of the Al³⁺ cation.³⁶

3. Additive engineering

3.1. Methods to add chloride additives

Many methods have been developed to incorporate the Cl additives into the fabrication process of PSCs. Additive engineering is one of the most efficient way to do that, in which a certain amount of Cl additive is added into the precursor solution, which is called a one-step strategy, as shown in Fig. 6a. Besides that, MACl can also be used to treat the PbI₂ film formed in the two-step process, where the role of the MACl is more complex due to the solid liquid interface reactions (Fig. 6b). It was found that the Cl treatment was effective even after the perovskite film was formed. Fig. 6c shows that the pristine MAPbI₃ thick film had randomly oriented crystal grains. After MACl solution was deposited onto the top of the MAPbI₃, an oriented intermediate phase would be formed. During the annealing process, the cubic perovskite with the (100) crystal plane was formed at the top surface. After the annealing process, the perovskite film was completely translated into tetragonal perovskite phase with the oriented (110) crystal plane. The GIWAXS patterns show the crystallographic orientation propagation during the post treatment process.³⁷

Huang *et al.* recently studied the role of MACl and FAPbCl in a dynamic sequential process.³⁸ It was found that the Cl in general (regardless of the type of source and at which step it is added) led to a perovskite film with better surface coverage, narrow diffraction peaks, higher PL intensity and longer carrier lifetimes because of reduced trap density. They found that the quantity of added volatile Cl additive is important in the

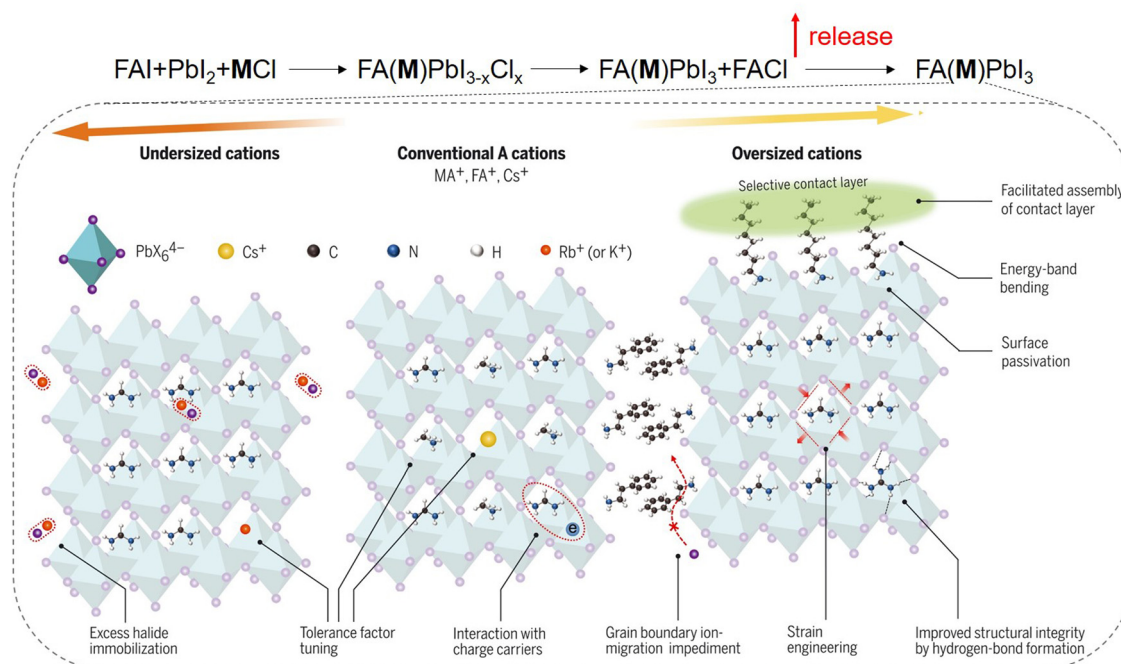


Fig. 5 Cation residues left by nonvolatile Cl additives. M is the cation with high boiling point. Inset is the possible location of the cations with increased size. Reproduced with permission.⁸ Copyright, 2022 AAAS.

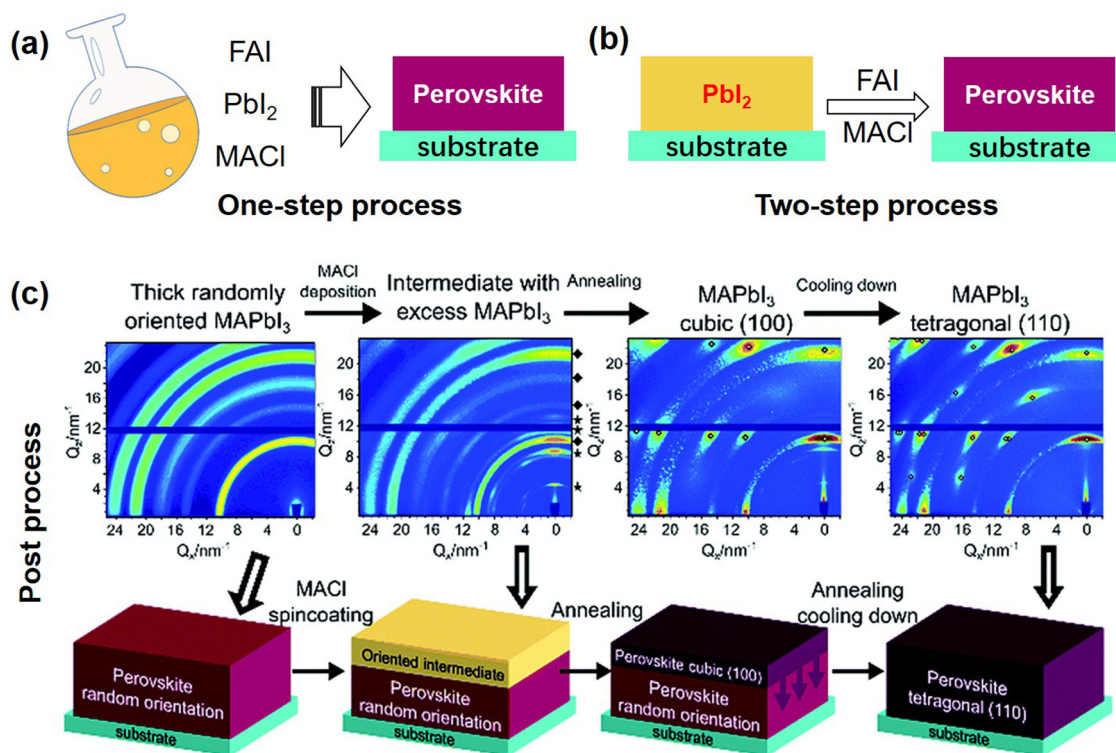


Fig. 6 Schematic of the methods to incorporate Cl additives in perovskite fabrication, the (a) one-step, (b) two-step and (c) post treatment process. Reproduced with permission.³⁷ Copyright, 2017 Royal Society of Chemistry.

resultant perovskite film quality. Therefore, the time when the Cl additive is added becomes important. If Cl is added in the second step, the high content and highly diffusive Cl induces

fast perovskite formation and results in excess PbI_2 residues, which leads to a disordered perovskite film with low crystallinity. FACl exacerbates this while MACl reduces the detrimental effect

Highlight

by slowing down perovskite formation. The authors suggested that MACl is preferred as the Cl source and the Cl should be added in the first step. The most Cl could get released in the second step.

3.2. Organic vs. inorganic

Typical solvents of perovskite precursors are DMF, DMSO and/or gamma-butyrolactone (GBL). Most of the organic Cl additives have a high solubility than that of their inorganic Cl additive counterparts. For example, Seok *et al.* added as high as 35 mol% of MACl and 15 mol% of FAcI to the FAPbI₃ precursor solution to modulate the crystal growth.²⁵ In comparison, the quantity of inorganic Cl additive is usually less than 5%. For example, Shekargoftar *et al.* added only 1% CdCl₂ into the MAPbI₃ solution and observed a PCE increase from 12.17% to 15.28% due to decreased charge carrier recombination.³⁹ Dai *et al.* added 3% zinc chloride (ZnCl₂) into the MAPbI₃ solution, and obtained an efficiency of 18.2%, while the control device only obtained an efficiency of 16.4%.⁴⁰ It should be noted that the amount of Cl is important in the crystallization modulation. Organic Cl additive in general is very effective in promoting crystal growth. The obtained crystal grain size is usually much larger than that of the pristine perovskite film (Fig. 2a and b). By contrast, the Cl from the inorganic additives was usually not sufficient for crystal growth modulation. Therefore the grain size was increased slightly (Fig. 2e and f), whose role is more related to the bulk defect passivation. Recently, You *et al.* incorporated 5% rubidium chloride (RbCl) into FAPbI₃ solution and got a high efficiency of 25.6%.⁴¹ It was found that excess PbI₂ existed in their two-step process, which is shown as a gray

area in Fig. 7a. Fig. 7b shows that RbCl could convert the PbI₂ into a secondary phase (PbI₂)₂RbCl, which was mainly located at the surface of FAPbI₃ (Fig. 7c). The new peak at around 11.4° confirmed the existence of the secondary phase (Fig. 7d). Because of the inactive secondary phase, the device exhibited an excellent stability and high efficiency due to the implementation of RbCl. Fig. 7e shows that the PSC with RbCl additive has a maximal PCE of 26.1% extracted from the *J*-*V* scanning, and 25.7% from the steady output for 1200 s. Furthermore, the device with RbCl additive has better stability under the temperature of 85 °C (Fig. 7f).

3.3. Volatile vs. nonvolatile

The volatilizing capability of Cl additives is dependent on their boiling point, which means that almost all inorganic Cl additives are nonvolatile, while partially organic Cl additives (>190 °C) can also be regarded as nonvolatile, see Table 1 for the boiling point comparison. It is expected that a low boiling point organic Cl additive, for instance MA⁰ (−6.3 °C), will get released easily. However, for the FA⁰ (46.3 °C), PA⁰ (47.8 °C), and methylenediammonium (MDA⁰, 82.6 °C), the release rate is lower and thus the crystal growth route is different. For the guanidinium chloride (GACl), it has a critical boiling point of 132.9 °C (GA⁰) that is close to the typical annealing temperature of FAPbI₃ or MAPbI₃. The role of this kind of organic Cl additive is complex because cation can be left behind or get released depends on the annealing time and temperature. As for high boiling point organic Cl additives, such as the DA⁰ (265 °C), AI⁰ (229 °C) and PEA⁰ (196.5 °C), the anion will be partially or fully

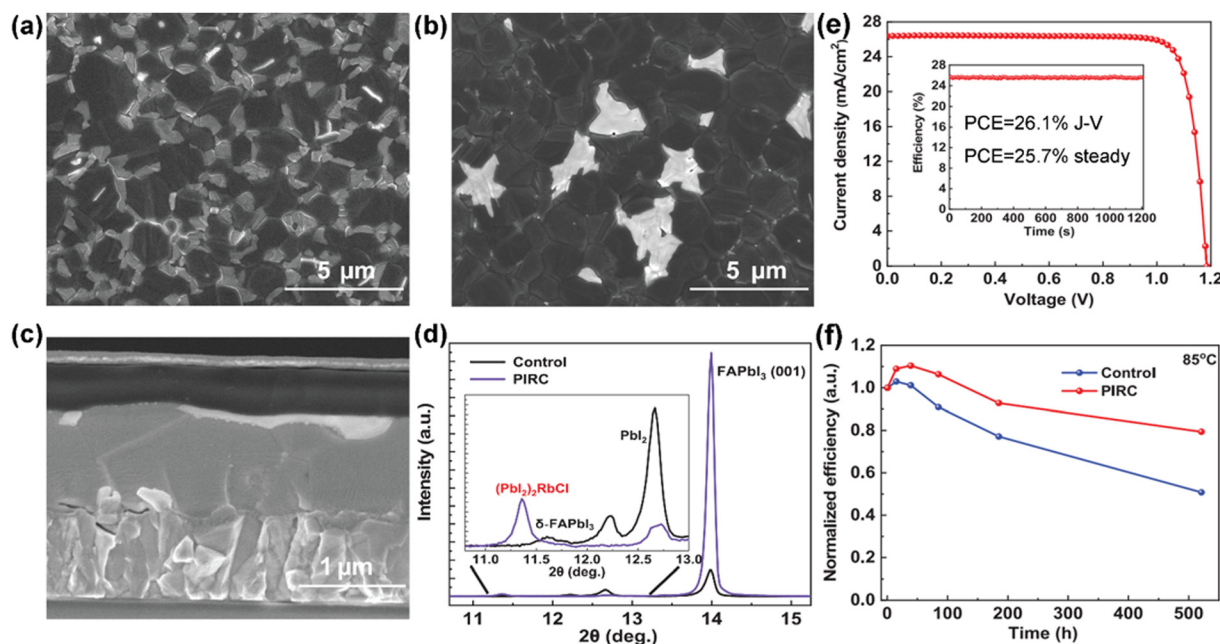


Fig. 7 Microstructures and morphology of the secondary-phase (PbI₂)₂RbCl on the perovskite film. (a) and (b) SEM of the perovskite films without RbCl and with 5% RbCl, respectively. Scale bar is 5 μm. (c) Cross-sectional SEM image of a completed device using RbCl doped perovskite. Scale bar is 1 μm. (d) XRD of perovskite without RbCl and with 5% RbCl. The performance of PSCs with RbCl additive, the PCE (e) and long-term stability (f). Reproduced with permission.⁴¹ Copyright, 2022 AAAS.

left as residues, whose role is dependent on the size of the cation and the chemical groups.

3.4. Cation size

Geometry size of the cations has a profound effect on the role of nonvolatile Cl additives. The conventional cations Cs^+ , MA^+ and FA^+ have a cation size of 167 pm, 217 pm and 253 pm, respectively, which results in a tolerance factor in the range of 0.8–1.0. If the undersized cations, such as Li^+ (76 pm),⁴³ Na^+ (95 pm),⁴⁴ K^+ (138 pm)⁴⁵ and Rb^+ (152 pm),⁴¹ are incorporated into the crystal lattice, the tolerance factor will be lowered. Furthermore, the ion migration will be greatly suppressed by the small halide cations. On the other hand, if the oversized cations, such as the MDA^{2+} (262 pm), DMA^+ (273 pm), EA^+ (274 pm), acetamidinium (Ace^+ , 277 pm) and GA^+ (278 pm), are introduced, the tolerance factor will be increased. The inset in Fig. 1 shows the relationship between the cation size and the

tolerance factor. In our recent study, we added a nonvolatile organic Cl additive AICI into the FAPbI_3 solution. It was found that a bottom-up 2D/3D perovskite heterojunction was formed at the buried interface due to the large cation size (742 pm). Fig. 8a shows the schematic of the 2D phase induced by the AICI additive. The cross section SEM shows the device structure and the high-resolution transmission electron microscopy (HRTEM) identified the coexistence of the 2D phase and the 3D phase (Fig. 8b), which has a crystal inter-layer space of 21.8 Å and 6.2 Å, respectively. The crystal grain size was increased slightly because the amount of AICI additive is small.³⁶ Because of the excellent passivation effect from the 2D phase, we obtained a PCE as high as 25.12% and high operation stability under damp-heat conditions (Fig. 8f). Compared to the AI^+ cation (742 pm), the cation size of DMA^+ (273 pm) is much smaller and volatile (boiling point of 7 °C). Therefore, the adding amount can be as high as 80%. Snaith *et al.* found that

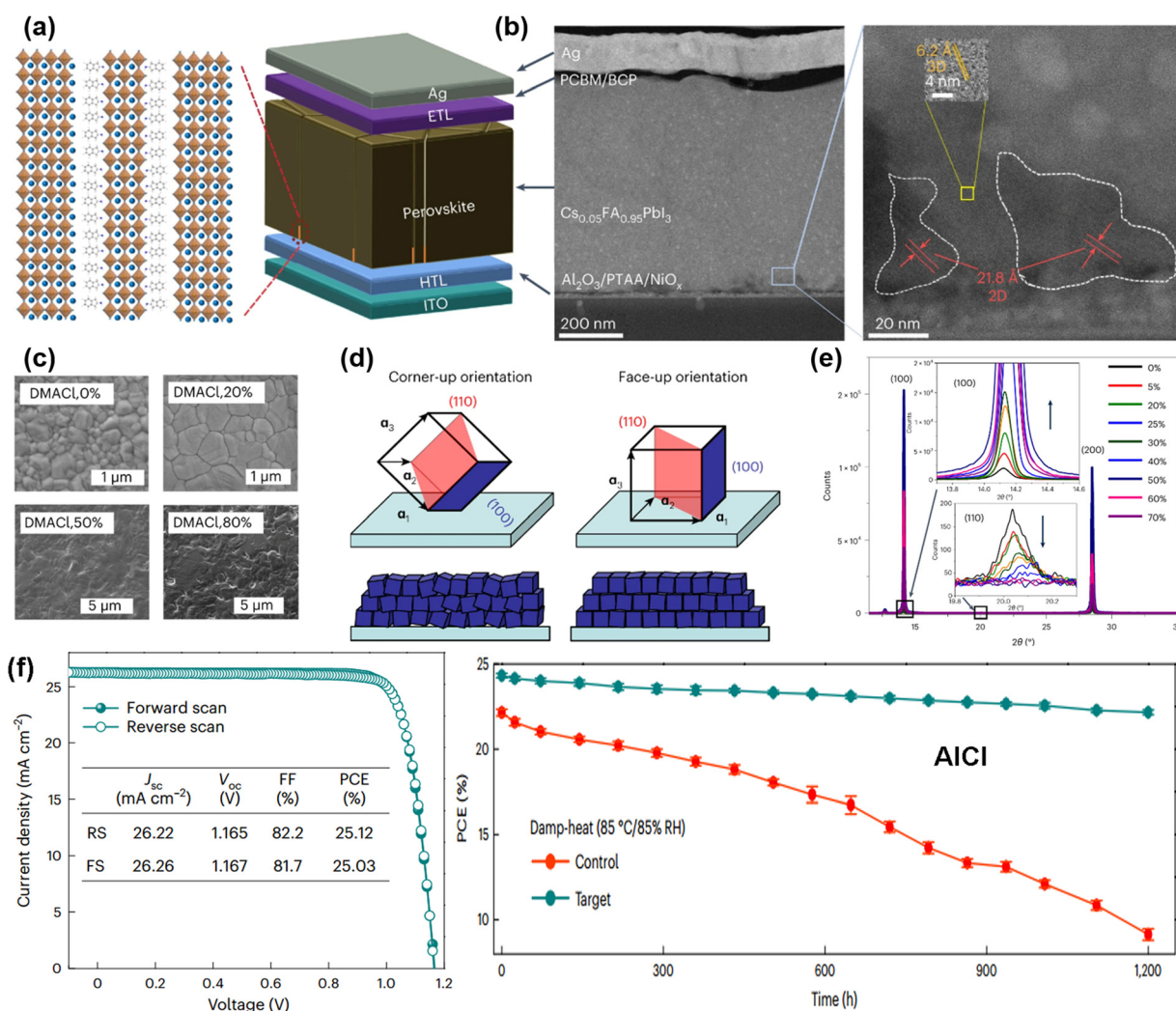


Fig. 8 (a) Schematic of the 2D perovskite phase induced by the AICI additive. (b) SEM and the HRTEM of the 2D phase at the buried interface. Reproduced with permission.³⁶ Copyright, 2023 Nature; (c) SEM images of $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$ with different DMACl concentrations. (d) Illustration of the crystal orientation ordering induced by DMACl. (e) XRD of the perovskite film after a full annealing step. The inset shows the 110 and 100 peak intensity evolution as a function of DMACl concentration increase. Reproduced with permission.⁴² Copyright, 2023 Nature. (f) PCE and long-term stability (85 °C/85% RH) of solar cell with AICI additive, Reproduced with permission.³⁶ Copyright, 2023 Nature.

Highlight

the incorporation of a high content of DMAcI can change the surface morphology of perovskite films. As shown in Fig. 8c, low content of the Cl additive led to a surface morphology with GBs, and high content of Cl additive resulted in a surface morphology without GBs. They proposed that the 110 corner-up orientation was preferred when the Cl content was low, while the 110 face-up orientation was formed when Cl content was high (Fig. 8d). This observation was further confirmed by the XRD intensity evolution as the content of Cl increases (Fig. 8e). The speed of crystallization should account for the surface morphology change. Cl additive can slow down the crystal growth, and thus the face-up orientation is preferred. It should be noted that the cation size, and volatile, organic and inorganic features of a Cl additive are strongly correlated with each other.

3.5. Cation residue location

Final location of the cation is dependent on the volatile capability and size of the Cl additives.²¹ In general, the inorganic Cl additives and organic ones with high boiling point will leave the cation as residues. According to the size of the cations, they are incorporated into the crystal lattice (small),⁴⁶ or accumulated at the grain boundaries (medium),⁴⁷ or form a layer at the buried interface or on the perovskite top surface (large).⁴⁸ One consideration for the large-sized cation residues is their conductivity. If a thick layer is formed at the buried interface or top surface, the series resistance will increase, which then leads to a decreased FF.³⁶ In such a way, the content of Cl additive with large cation size is in general low; however, low content of Cl in turn results in a weak control of the crystal nucleation and growth process.

3.6. Chemical group

Compared to the inorganic Cl additives, the rich choice of chemical groups of organic Cl additives could finely manipulate the nucleation and crystal growth process, or effectively passivate the defects and be helpful in 2D phase formation. At present,

more than 100 different kinds of organic Cl additives are available from the chemical companies, most of which have not been explored yet. Fig. 9 shows the chemical structures of typical organic Cl additives. They have different cation sizes, solubility, boiling point and chemical groups. It is well known that the uncoordinated Pb^{2+} is Lewis acid, which could be efficiently passivated by the electron rich chemical groups, such as the $-\text{NH}_2$, $-\text{S}$, and $-\text{O}$.^{49,50} Small alkyl molecules with passivation groups are frequently utilized as nucleation and crystal growth modulation agents due to their volatile capability and high solubility. Huang *et al.* have added the choline chloride into the FAPbI₃ precursor solution, and achieved an absolute efficiency increase of 1.8%.⁵¹ Further increase in the cation size from alkyl to phenyl groups and functionalizing them with the $-\text{O}$ groups could be an effective way from our viewpoint. It was found that not only the chemical groups themselves have effects on the passivation, but also the configuration of the molecules, the stereochemistry and chirality of the chemical groups can affect the interaction between perovskite and organic Cl additives.^{52,53} The stereomolecule 1-adamantylamine hydrochloride (ADAHCl) has been added into FAPbI₃ perovskite to further improve the performance.³¹ Furthermore, supramolecular complexes such as 2-hydroxypropyl β -cyclodextrin and 1,2,3,4-butane tetracarboxylic acid (HP β CD-BTCA) has been used to prevent lead leakage.⁵⁴ Similarly, the supramolecular daunorubicin hydrochloride might also find application in lead leakage prevention and crystal growth modulation due to the Cl anions. The chemical group engineering of small molecules has been proved to be rather effective in defect passivation for PSCs.^{55,56} It is expected that more and new organic Cl additives can be developed by precisely controlling the kind, number and spatial conformation of functional groups in the future.

3.7. Limitations of the Cl additive engineering

In the scalable PSC production process, such as the slot-die coating and spray coating process, the amount of solvent is

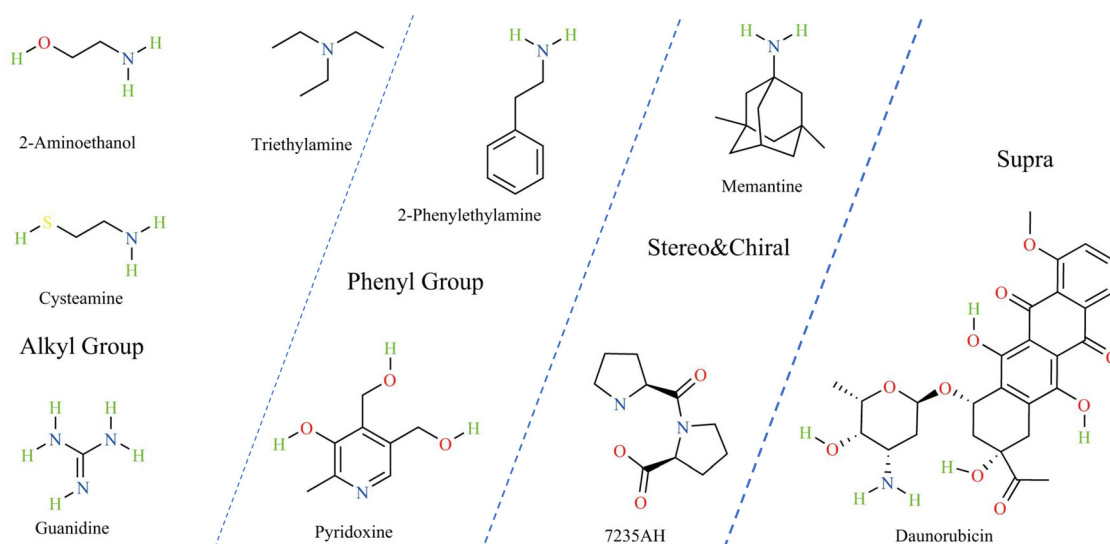


Fig. 9 Chemical groups that can be explored to tune the crystal growth, passivate defects and facilitate 2D phase formation.

around 2.5 ml m^{-2} for a 500 nm thick perovskite layer, which is equal to 3500 liters of solvent for a 1 GW factory, assuming module efficiencies of 15% and a high module yield.⁵⁷ Note that the solvents used in PSC manufacturing process are all small in comparison to analogous electronic technologies, for example the production of 30 000 17" LCD displays. DMF, DMSO and GBL are the most frequently used solvent for PSC production, where the DMF is the most dangerous and toxic solvent that draws the most concern.⁵⁸ As the Cl additives are added into the DMF/DMSO solvent, the halogenated solvent would become more problematic than before, therefore solvent recycling is needed in the GW industrial production. Furthermore, for some inorganic Cl additives, such as CdCl_2 and RbCl , which are toxic for the environment and human health, high reliability encapsulation is required to prevent it from leaking into environment.

4. Conclusions and outlook

As the commercialization of PSCs has moved forward in recent years, the modulation of nucleation and crystal growth over a meter scale area becomes one of the most important factors that determines the final performance of the solar panels. Cl additives have gained their reputation in crystal growth modulation due to their role of intermediate phase agents. Both the inorganic and organic Cl additives have been widely used in the PSC research community. The main role of organic Cl additives with a low boiling point is believed to facilitate crystal growth and gets released after the annealing process, whose solubility in a typical perovskite solution is high and a large amount of Cl additive is required for effective crystal growth modulation. In general, a high content of volatile organic Cl additives leads to an increased crystal grain size, which could reach the μm scale. In comparison, the role of inorganic Cl additives and high boiling point organic Cl additives is much more complex. Because of the nonvolatile features at the typical annealing temperature, small inorganic cations and large organic cations are left behind as residues, which can immobilize the excess halide, tune the structure tolerance factor, interact with charge carriers, impede ion migration at the grain boundary, release crystal strain, passivate defects, form hydrogen bonds, and align energy levels according to their sizes, chemical groups and locations. As the content of the inorganic Cl additives and high boiling point organic Cl additives is relatively low, the role of crystal modulation is much weaker. Therefore, it could be an effective approach to increase the solubility of high boiling point organic Cl additives by the functionalization of functional groups and cation engineering.

Although many studies related to Cl additives have been performed in recent years, there are still spaces that are worth exploring to further facilitate the development of this field and to expand new applications. Here are some of our viewpoints on the potential future directions on the application of Cl additives in PSCs:

(1) Elucidation of the mechanism. First-principles based calculations and *in situ* characterizations should be performed

to gain more insights into the role of volatile organic Cl additives.^{59–61} More efforts should be devoted to revealing the role of the nonvolatile inorganic Cl additives and the organic ones with high boiling point temperatures.

(2) Design and synthesis of new multifunctional organic Cl additives. The structural diversity of organic cations provides the research community with a wide choice of tools to passivate the bulk and interface defects beyond the crystal growth modulation provided by Cl anions. In other words, functionalizing the cations with active groups which can passivate defects, modulate perovskite crystallization, suppress ion migration and release residual stress is necessary for developing more effective multifunctional new organic Cl additives. Upon the rational design of the cations, we should simultaneously consider the solubility, volatile capability, chemical interaction and tolerance factor.

(3) Expand the application of organic Cl additives to inorganic PSCs. Hybrid organic–inorganic PSCs are the mainstream of high-efficiency solar cells. However, the poor stability under high temperature has become a critical issue for long-term operation. PSC research community should try to explore the application of organic Cl additives in inorganic PSC fabrication by tailoring the cation chemical groups.

(4) Synergistic use of the additives with Cl^- and other anions. The use of two or more additives has become a trend. The synergistic utilization of organic, inorganic, volatile, non-volatile, low boiling point and high boiling point Cl additives is worth exploring for high-quality perovskite films.

Overall, the success of Cl additives in high performance solar cells has gained extensive attention from the research community and in the industrial fields. We firmly believe that the Cl additives will become a key participant in the additive engineering of perovskite solar panels. Moreover, the Cl additives can make bigger contributions to the development of PSCs and other perovskite optoelectronic devices.

Conflicts of interest

There are no conflicts to declare.

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